Eduard Zamfir

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Education

Ph.D. Computer Science, University of Würzburg *Aug 2023 – Present*
- Expected graduation: August 2027
- Focus: Computer Vision, Machine Learning
- Advisor: Prof. Radu Timofte

M.Sc. Computational Engineering, Technical University of Darmstadt *Apr 2020 – May 2022*
- Graduated with honors (Grade: 1.19, GPA 4.0 equivalent)
- Focus: Computer Vision, Machine Learning
- Advisor: Prof. Stefan Roth

M.Sc. Mechanical Engineering, Technical University of Darmstadt *Oct 2017 – Mar 2020*
- Completed 114/120 ECTS; 46 credits transferred to M.Sc. Computational Engineering

B.Sc. Mechanical Engineering, Technical University of Darmstadt *Oct 2013 – Oct 2017*
- Focus: Mechatronics, Automotive Engineering

Publications / Preprints

1. **E. Zamfir**, C. Reisswig, Z. Wu, Y. Xian, R. Timofte, "Elastic Token Compression for Pixel-Space Diffusion Transformers", arXiv, 2026.
2. O. Elezabi, **E. Zamfir**, Z. Wu, R. Timofte, "Language-Free Generative Editing from One Visual Example", *CVPR*, 2026.
3. Y. Tan, J. Shao, **E. Zamfir**, R. Li, Z. An, C. Ma, D. Paudel, L. van Gool, R. Timofte, Z. Wu, "What You Have is What You Track: Adaptive and Robust Multimodal Tracking", *ICCV*, 2025
4. Y. Tan, Z. Wu, Y. Fu, Z. Zhou, G. Sun, **E. Zamfir**, C. Ma, D. Paudel, L. van Gool, R. Timofte, "XTrack: Multimodal Training Boosts RGB-X Video Object Trackers", *ICCV*, 2025
5. **E. Zamfir**, Z. Wu, N. Mehta, Y. Tan, D. Paudel, Y. Zhang, R. Timofte, "Complexity Experts are Task-Discriminative Learners for Any Image Restoration", *CVPR*, 2025.
6. J. Li, Z. Wu, **E. Zamfir**, R. Timofte, "ReCap: Better Gaussian Relighting with Cross-Environment Captures", *CVPR*, 2025.
7. **E. Zamfir**, Z. Wu, N. Mehta, Y. Zhang, R. Timofte, "See More Details: Efficient Image Super-Resolution by Experts Mining", *ICML*, 2024.
8. **E. Zamfir**, Z. Wu, N. Mehta, D. Paudel, Y. Zhang, R. Timofte, "Efficient Degradation-aware Any Image Restoration", arXiv, 2024.
9. B. Ren, **E. Zamfir**, Z. Wu, Y. Li, Y. Li, D. Paudel, R. Timofte, M.H. Yang, N. Sebe, "Restore Anything Model with Efficient Degradation Adaptation", arXiv, 2024
10. Z. Fang, A. Ignatov, **E. Zamfir**, R. Timofte, "SQAD: Automatic Smartphone Camera Quality Assessment and Benchmarking", *ICCV*, 2023.
11. **E. Zamfir**, M. Conde, R. Timofte, "Towards Real-Time 4K Image Super-Resolution", *CVPRW*, 2023.
12. S. Bahmani*, O. Hahn*, **E. Zamfir***, N. Araslanov, D. Cremers, and S. Roth, "Semantic Self-Adaptation: Enhancing Generalization with a Single Sample", *TMLR*, 2023.

Research Experience

Research Intern, Huawei Research Zurich *Sep 2026 – Feb 2027*
- Advisor: Anton Obukhov, Dengxin Dai

Researcher, Computer Vision Lab, University of Würzburg *Aug 2022 – Aug 2023*
- Real-time Image Super-Resolution, Camera Quality Assessment
- Advisor: Radu Timofte

Master Thesis, Visual Inference Lab, TU Darmstadt *Nov 2021 – May 2022*
- Exploiting non-local dependencies for image restoration using attribution priors
- Advisors: Robin Hesse, Prof. Stefan Roth

Student Researcher, Visual Inference Lab, TU Darmstadt *Sep 2020 – Sep 2021*
- Domain generalization for semantic segmentation

- Advisors: Nikita Araslanov, Prof. Stefan Roth

Student Researcher, Fraunhofer LBF *Apr 2019 – Apr 2020*
- HiL-testbench for driving dynamics simulations
- Advisors: Riccardo Bartolozzi, Prof. Tobias Melz

Bachelor Thesis, Institute for Mechatronic Systems, TU Darmstadt *Apr 2017 – Oct 2017*
- Simulation of hybrid energy storage systems for industrial applications
- Advisor: Prof. Stefan Rinderknecht

Work Experience

Intern, Porsche Engineering Group *Aug 2018 – Feb 2019*
- Driving Performance Team: Software engineering for driving dynamics simulation

Academic Service

Reviewer: CVPR, ICCV (Outstanding Reviewer 2025), NeurIPS (Top Reviewer 2026), ICLR, ICML, AAAI, TPAMI, IJCV, WACV, TIP, ACM Multimedia

Conference Workshops: Co-Organizer 2nd AIGENS Workshop, ICCV 2025; Co-Organizer NTIRE Challenge on Efficient Super-Resolution, CVPR 2024; Co-Organizer NTIRE Challenge on Real-Time 4K Super-Resolution, CVPR 2023

Teaching Assistant: Computer Vision, Image Processing and Computational Photography

Academic Supervision

Student Project: Roman Kochnev, University of Würzburg *Ongoing*
- Topic: Dynamic Vision Models for Image Restoration
- Co-Supervisors: Prof. Radu Timofte

Master Thesis: Tobias Brandner, University of Würzburg *2024*
- Topic: Real-Time Rendering Super Resolution with Unreal Engine 5
- Co-Supervisor: Prof. Radu Timofte, Nancy Mehta

Technical Skills

Programming: Python (PyTorch, NumPy, OpenCV), Matlab, Git, LaTeX, Bash, Linux

Machine Learning: Generative Models (Diffusion Models, Diffusion Transformers), Image Generation & Editing, Image Restoration, Super-Resolution, Vision Transformers, Mixture-of-Experts, Token Compression, Semantic Segmentation, Multimodal Tracking

Languages: German - Native, **English** - Fluent, **Romanian** - Native, **French** - Basic (UNIcert B1)